Sardar Vallabhbhai National Institute of Technology (SVNIT) Surat
Department of Artificial Intelligence
5th Sem , B.Tech. (AI) (2026-27)

Information Security and Cryptography - AI 331
Lab Assignment 4

Scenario

A ground control station needs to transmit a fast, tactical command to a field drone using a 5 X 5 Playfair Cipher.

- **Key:** DRONE
- **Plaintext:** ATTACK AT THE STATION

Tasks

Write a code to :

1. Construct the 5 X 5 key matrix using the key DRONE.
2. Format the plaintext into valid 2-letter digraphs (handle repeated letters and odd lengths using 'X').
3. Encrypt the digraphs to produce the final ciphertext.
4. Decrypt the ciphertext back to verify the plaintext.